

Automatic Phenotyping from Clinical Notes using MIMIC-IV

The George Washington University

CSCI 6907: NLP for Health

Prof. Ayah Zirikly

10th December 2025

Presented By :-

Akash Raghavendra - G41868923

Nidhi Naidu - G38443562

Sai Srinivas - G20957666

ABSTRACT

The goal of this project is to examine whether unstructured clinical notes can be used to automatically identify patients with heart failure and diabetes using the MIMIC-IV dataset. We focus on discharge summaries and fine-tune a transformer-based model, BioClinical ModernBERT, to predict the presence of these phenotypes using labels derived from ICD-10 codes. To understand where disease-specific signals originate, we include only clinically relevant note sections rather than using full documents. The model was evaluated against a baseline transformer without fine-tuning. The baseline model showed limited ability to identify diabetes and was unable to reliably detect heart failure. After fine-tuning, performance improved substantially for both conditions, indicating that the model was able to capture clinically meaningful signals.. These findings imply that section-based selection can facilitate the efficient prediction of chronic conditions from clinical narratives alone and that discharge summaries contain enough information for automated phenotyping.

INTRODUCTION

Electronic health records contain both structured fields, such as ICD codes, and rich narrative descriptions written by clinicians. While diagnosis codes are widely used for research and quality measurement, they can suffer from incorrect entry, documentation inconsistencies, and variation in coding practice. In contrast, discharge summaries often reflect the full clinical picture, including presenting symptoms, disease evolution, comorbidities, and treatment decisions. This makes them a promising source of information for identifying disease phenotypes directly from text.

This project explores whether discharge summaries alone can be used to identify two common conditions - heart failure and diabetes mellitus in the MIMIC-IV database. By limiting the input to selected sections of the note, we aim to understand not only the performance of transformer models on this task, but also how much useful signal is contained in the narrative parts of the document.

Research Questions

Our work is guided by two questions:

RQ1. Can unstructured discharge summaries accurately identify patients with heart failure and diabetes when compared to ICD-based phenotyping?

RQ2. Which narrative sections within discharge summaries carry the strongest predictive signals for disease classification?

RELATED WORK

Zhang et al.[\[1\]](#) demonstrated that extracting and integrating phenotypic concepts from clinical notes with structured EHR data substantially improves ICU outcome prediction, outperforming models that use only structured features. Their results show that unstructured notes capture vital clinical context and predictive information that tabular data alone often misses. This relates to our project's focus on leveraging NLP to extract phenotypes from clinical notes for more accurate patient identification and risk modeling. Overall, the study validates the value of unstructured EHR data and strengthens our approach of combining textual and structured sources for clinical prediction.

Yang et al.[\[2\]](#) demonstrated that transformer based models like PhenoBCBERT and PhenoGPT outperform traditional ontology-driven tools in recognizing clinical phenotypes from real world notes, particularly capturing novel and context-dependent terms. Their results highlight how BERT and GPT architectures can effectively extract nuanced phenotype information even under limited training data conditions. This relates to our project's focus on leveraging advanced transformers for robust, scalable phenotype extraction from unstructured notes. Their analysis also aligns with our interest in balancing model accuracy and data efficiency in clinical NLP pipelines.

Munzir et al.[\[3\]](#) directly compared GPT-4, a deep learning spaCy pipeline and a hybrid machine learning method for extracting neurological phenotypes from physician notes, finding that GPT-4 consistently outperformed the other approaches in both accuracy and efficiency. The study highlighted GPT-4's ability to handle complex language, class imbalance and varied data representations while reducing manual tuning and annotation requirements. This evidence reinforces the value of advanced transformer-based models in clinical phenotyping tasks. Their findings strongly relate to our project's focus of adopting large language models for high throughput, accurate phenotype extraction from unstructured clinical texts.

DATA

We use the MIMIC-IV database, a publicly available, de-identified dataset released by the MIT Laboratory for Computational Physiology[4]. The dataset includes extensive hospital stay information, such as demographics, diagnoses, laboratory values, and narrative clinical notes.

1	Patients table	demographic details
2	Admissions table	hospitalization-level data
3	Diagnoses_ICD table	ICD-10 diagnosis codes used as phenotype labels
4	MIMIC-IV Note (Discharge Summaries)	unstructured clinical text containing detailed patient summaries

METHODS

Phenotype Labeling

Generated binary labels for two conditions using ICD-10 diagnosis codes. A patient admission was labeled positive (1) if any of the associated diagnosis codes matched with:

Heart Failure: Code - I50.

Diabetes Mellitus: Codes starting with E08, E09, E10, E11, or E13.

Ensured the data quality by removing:

- Records with missing or empty discharge summary text.
- Duplicate hospital admissions (keeping the first occurrence per hadm_id).

Stratified Sampling

To address the inherent class imbalance in clinical data, we utilized a stratified sampling technique and divided the dataset into four distinct subgroups:

- *Heart Failure only*
- *Diabetes only*
- *Combined (Both Heart Failure & Diabetes)*
- *Neither*

Balanced the dataset by determining the count of each positive subgroup (HF only, Diabetes only, Both) and sampling an equivalent number of negative examples from the "Neither" pool. Once a negative sample was used for one subgroup, it was removed from the pool to prevent overlapping. The final dataset was shuffled and consisted of “128,776” unique records.

```
Sampled dataset size: 128776
Label distribution (sum of 1s):
heartfailure_label    23451
diabetes_label        56976
dtype: int64
```

Text Preprocessing

Clinical notes typically contain administrative metadata irrelevant to phenotyping. A regex-based segmentation is used to extract only the sections most indicative of a patient's clinical status. The extracted sections are: ‘History of present illness’, ‘Past Medical History’, ‘Hospital course’, ‘Discharge Diagnosis’, ‘Discharge medications’. The text from these sections were lowercased and then stripped of whitespace, underscore runs, excessive new lines and bracketed content.

Data Splitting and Tokenization

The processed dataset was split into training (70%), validation (15%), and testing (15%) sets. Crucially, we used the stratify parameter based on the combination of labels to ensure the distribution of labels remained consistent across all splits.

The tokenizer of the BioClinical ModernBERT[\[6\]](#) model is utilized. The text was padded and truncated to a maximum sequence length of 4096 tokens, generating PyTorch tensors for input_ids and attention_mask.

Training Setup

We fine-tuned the BioClinical ModernBERT model and the training was performed on a single NVIDIA A100 GPU (80GB VRAM). The batch size was 112, ran for 5 epochs and the optimizer used was AdamW optimizer.

Data Validation

Prior to evaluation, strict data validation checks were performed to ensure the integrity of the input tensors:

Structural Inspection: Verifying that saved objects contained only numeric tensors and no residual metadata.

PHI & Raw Text Detection: Automated scans were run to ensure no accidental raw text strings or Potential Health Information (PHI) were preserved in the tokenized files, adhering to data privacy standards.

Tensor Summary Audit: Statistical inspection of the tensors was conducted to confirm they represented valid token IDs within the model's vocabulary range.

EVALUATION AND RESULTS

Precision: To measure the proportion of predicted positive phenotypes that were correct (Positive Predictive Value).

Recall: To measure the proportion of actual positive phenotypes that were correctly identified. In clinical screening, high recall is often prioritized to minimize missed diagnoses.

F1-Score: The harmonic mean of precision and recall, providing a balanced single-metric view of performance.

Baseline Model

To understand the value added by the fine-tuning process, we established a baseline using the pre-trained BioClinical-ModernBERT model without any task-specific weight updates. This baseline evaluation has been done on the test set using the exact same preprocessing and tokenization pipeline.

Results

We compared the performance of the pre-trained BioClinical ModernBERT model against our fine-tuned version on the held-out test set.

Baseline Performance: The pre-trained BioClinical ModernBERT model, when used directly for the classification task without fine-tuning, failed to capture the complexity of the phenotypes, which is reflected in the performance metrics shown below.

	precision	recall	f1-score	support
Heart_Failure	1.00	0.00	0.00	3518
Diabetes	0.44	0.30	0.36	8547
micro avg	0.44	0.21	0.28	12065
macro avg	0.72	0.15	0.18	12065
weighted avg	0.61	0.21	0.25	12065
samples avg	0.13	0.11	0.12	12065

Fine-Tuned Performance: After fine-tuning, the model demonstrated substantial improvements across all metrics.

	precision	recall	f1-score	support
Heart_Failure	0.69	0.56	0.61	3518
Diabetes	0.68	0.88	0.77	8547
micro avg	0.68	0.78	0.73	12065
macro avg	0.69	0.72	0.69	12065
weighted avg	0.68	0.78	0.72	12065
samples avg	0.40	0.40	0.39	12065

DISCUSSION AND ANALYSIS

To contextualize these results, we compared our model's performance against established benchmarks for phenotyping classification tasks:.

Our fine-tuned model (Macro F1 0.69) outperformed the standard BioClinicalBERT[5], which typically achieves an F1-score of approximately 0.59 on similar biomedical classification tasks.[5]

Our performance is competitive with state-of-the-art retrieval-based models like MedTE (F1~0.72)[5] and Nomic AI Embed Text (F1 ~0.69)[5], validating the benefit of using the ModernBERT architecture with its extended context window for long clinical notes. We also ran the same experiment on the ModernPubMedBERT and its results are shown below:

Modern-Pubmedbert baseline Eval:

	precision	recall	f1-score	support
Heart_Failure	0.19	0.72	0.30	3518
Diabetes	0.45	0.93	0.61	8547
micro avg	0.34	0.87	0.48	12065
macro avg	0.32	0.82	0.45	12065
weighted avg	0.37	0.87	0.52	12065
samples avg	0.33	0.44	0.36	12065

Modern-Pubmedbert Finetuned Eval:

	precision	recall	f1-score	support
Heart_Failure	0.70	0.57	0.63	3518
Diabetes	0.73	0.78	0.76	8547
micro avg	0.73	0.72	0.72	12065
macro avg	0.72	0.67	0.69	12065
weighted avg	0.72	0.72	0.72	12065
samples avg	0.36	0.36	0.35	12065

ETHICS

The proposed project will be conducted in alignment with the ethical, privacy, and data usage standards set forth for the MIMIC-IV dataset.

- **Data Privacy & De-identification:** MIMIC-IV is already de-identified under HIPAA Safe Harbor (with PHI removed or replaced). We will only use this de-identified data and will not attempt re-identification.
- **Data Usage Agreement:** All team members have or will complete the required CITI training. We will comply with the data usage agreement and restrict dataset use to this course project.
- **Interpretability & Use:** We will apply interpretability techniques (e.g., attention or feature importance) to understand model decisions. The system is for research only and not intended for direct clinical deployment.

LIMITATIONS

Despite these promising results, several limitations must be acknowledged. First, the model was fine-tuned and evaluated on data from a single institution (MIMIC-IV), which limits claims regarding its generalizability to EHR systems from other hospitals or demographic populations. Second, due to computational constraints, we were unable to conduct extensive hyperparameter optimization or cross-validation experiments. Finally, the system is currently a standalone prototype; its performance and utility within a live, real-time clinical decision support workflow remain to be validated.

CONCLUSION AND FUTURE WORK

This study successfully demonstrated the efficacy of using fine-tuned, long-context transformer models for automating patient phenotyping from unstructured clinical notes. By leveraging the BioClinical ModernBERT architecture, we addressed a critical limitation in traditional clinical NLP: the inability to process lengthy discharge summaries without truncation. Our experimental results highlight that while pre-trained biomedical models possess domain knowledge, they are insufficient for accurate phenotyping. Fine-tuning proved to be the decisive factor, boosting the Macro F1-score from ‘0.18’ to ‘0.69’ and enabling the accurate identification of complex conditions like Heart Failure and Diabetes. Furthermore, our analysis revealed that the

BioClinical ModernBERT performs well after fine-tuning, suggesting that the training process effectively adapts distinct pre-training foundations to the specific nuances of the MIMIC-IV dataset. Future work will focus on cross-institutional validation to ensure robustness and the development of lightweight deployment strategies to integrate these powerful phenotyping tools directly into clinical workflows

CONTRIBUTIONS

Akash Raghavendra - Literature review, model evaluation and model development.

Nidhi Naidu - Literature review, data extraction and feature representation

Sai Srinivas G - Literature review, data extraction, data preprocessing, comparative analysis

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